IQC 17/5/2024

$Ax = b \qquad A \in \mathbb{C}^{N \times N}, \quad \det A \neq 0$

$x = A^{-1} b$

In the classical case if A is dense $O(N^3)$

if A is sparse with iterative methods (CG), if s is the average number of nonzeros per row $O(N s \, \mu(A) \log(\frac{1}{\varepsilon}))$

$\mu(A) = \|A\| \cdot \|A^{-1}\|$

computed solution $\tilde{x}$: $\|\tilde{x} - x\| \leq \varepsilon$

HHL → Harrow – Hassidim – Lloyd

$Ax = b$

$|b\rangle = \frac{b}{\|b\|} \qquad \frac{1}{\|b\|} A x = |b\rangle$

The computed solution will be encoded in a quantum state

$$|x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|} = \frac{x}{\|x\|}$$

the normalization constant should be recovered independently.

A is hermitian $\qquad A = A^{+}$

$$\tilde{A} = \begin{bmatrix} 0 & A^{+} \\ A & 0 \end{bmatrix} \quad \text{is hermitian}$$

$$\tilde{A} \begin{bmatrix} |x\rangle \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ |b\rangle \end{bmatrix}$$

$$\tilde{A} = |1\rangle\langle 0| \otimes A + |0\rangle\langle 1| \otimes A^{+}$$

$$\begin{bmatrix} x \\ 0 \end{bmatrix} = |0\rangle \otimes |x\rangle \qquad \begin{bmatrix} 0 \\ |b\rangle \end{bmatrix} = |1\rangle \otimes |b\rangle$$

QLSP: Quantum linear system problem

Given A hermitian and $|b\rangle$ find $|\tilde{x}\rangle$ such that

$$\| |\tilde{x}\rangle - |x\rangle \|_2 \leq \varepsilon \qquad |x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|}$$

Since A is hermitian, A is diagonalizable by means of unitary transf.

$\{|v_0\rangle, |v_1\rangle, \ldots |v_{N-1}\rangle\}$ is a orthonormal basis of eigenvectors

$$\langle v_j | v_k \rangle = \begin{cases} 0 & \text{if } j \neq k \\ 1 & \text{se } j = k \end{cases}$$

$$A|v_j\rangle = \underbrace{\lambda_j}_{\text{real}} |v_j\rangle \qquad j = 0, \ldots N-1$$

$$A = V D V^{+} \qquad V = [\,|v_0\rangle \, |v_1\rangle \ldots |v_{N-1}\rangle\,]$$

$Ax = |b\rangle \qquad x = A^{-1}|b\rangle$

$$x = (VDV^\dagger)^{-1}|b\rangle = VD^{-1}\underbrace{V^\dagger|b\rangle}$$

$$|b\rangle = \sum_{i=0}^{N-1} \beta_i |v_i\rangle \qquad \{|v_0\rangle, \dots |v_{N-1}\rangle\} \text{ is a basis}$$

$$\beta_i = \langle v_i|b\rangle$$

$$x = \sum_{i=0}^{N-1} \frac{\beta_i}{\lambda_i} |v_i\rangle$$

To encode matrix A in a quantum circuit we can work with

$$U = e^{2\pi i A} \triangleq V e^{2\pi i D} V^\dagger \qquad e^{2\pi i D} = \begin{bmatrix} e^{2\pi i \lambda_0} & & 0 \\ & e^{2\pi i \lambda_1} & \\ 0 & & \ddots \; e^{2\pi i \lambda_{N-1}} \end{bmatrix}$$

- U is unitary if A is hermitian
- $U|v_j\rangle = e^{2\pi i \lambda_j}|v_j\rangle$

Assume that each λ_j is representable exactly with d-bits, For simplicity consider the case where

$$0 < \lambda_0 \le \lambda_1 \le \dots \le \lambda_{N-1}$$

$$QPE(U, |0\rangle^{\otimes d} \otimes |v_j\rangle) = |\lambda_j\rangle \otimes |v_j\rangle$$

$$QPE\big(U, \; |0\rangle^{\otimes d} \otimes \underbrace{|b\rangle}_{\sum \beta_j |v_j\rangle}\big) = \sum_{j=0}^{N-1} \beta_j \, |\lambda_j\rangle \otimes |v_j\rangle$$

$$x = A^{-1}|b\rangle = \sum_{j=0}^{N-1} \frac{\beta_j}{\lambda_j} |v_j\rangle$$

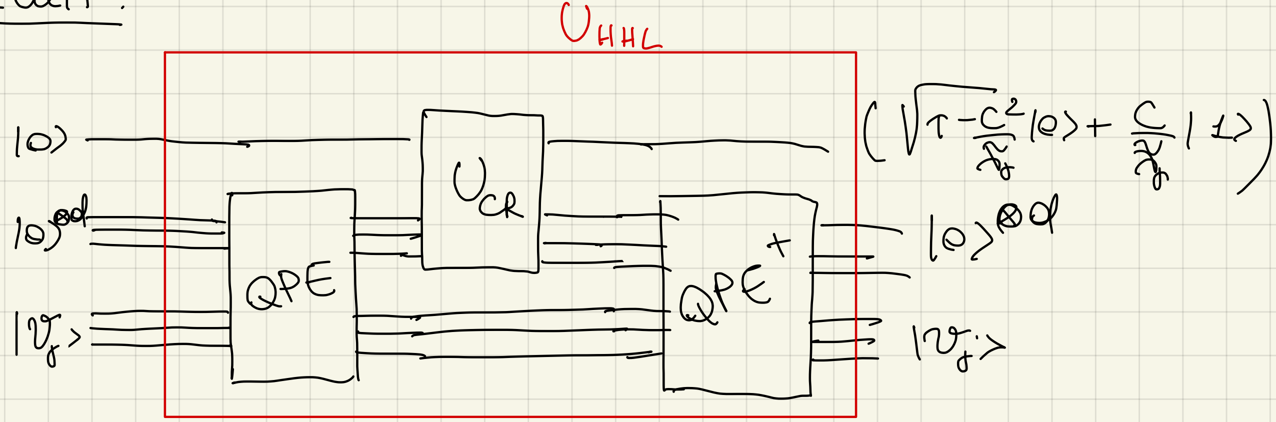
To compute $\frac{1}{\lambda_j}$ from $|\lambda_j\rangle$ we add an ancilla qubit and we perform a rotation controlled by the d-qubit register

$$U_{CR}\big(\underbrace{|0\rangle}_{\text{ancilla}} |\lambda_j\rangle\big) = \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j}}\, |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle\right) |\lambda_j\rangle$$

$\tilde{\lambda}_j$ approximates λ_j

C is a normalization parameter such that the amplitudes are in modulo less than 1.

<u>HHL circuit:</u>



$$U_{HHL}\left(|0\rangle|b\rangle\right) = U_{HHL}\left(\sum_{j=0}^{N-1} \beta_j |0\rangle|v_j\rangle\right) = \sum_{j=0}^{N-1} \beta_j U_{HHL}\left(|0\rangle|v_j\rangle\right)$$

$$= \sum_{j=0}^{N-1} \beta_j \left(\sqrt{1-\frac{C^2}{\tilde{\lambda}_j}}|0\rangle + \frac{C}{\tilde{\lambda}_j}|1\rangle\right)|v_j\rangle =$$

$$= \sum_{j=0}^{N-1} \beta_j \sqrt{1-\frac{C^2}{\tilde{\lambda}_j}}|0\rangle|v_j\rangle + C|1\rangle \underbrace{\sum_{j=0}^{N-1} \frac{\beta_j}{\tilde{\lambda}_j}|v_j\rangle}_{\approx |\tilde{x}\rangle}$$

Then we measure the ancilla qubit. If the outcome is $|0\rangle \rightsquigarrow$ we throw away the result we keep instead the state vector when the outcome is 1.

$$\tilde{x} = \sum_{j=0}^{N-1} \frac{C\beta_j}{\tilde{\lambda}_j}|v_j\rangle \quad \text{is not normalized}$$

$$\text{prob}(1) = \left\| \sum_{j=0}^{N-1} \frac{C}{\lambda_j}\beta_j |v_j\rangle \right\|^2 = C^2 \| A^{-1}|b\rangle \|^2$$

is the normalization parameter

For estimating such a probability we need to repeat the circuit many times.

If $C \approx \lambda_0$ the probability is maximized.

computational cost

$$\text{prob}(1) = \|\tilde{x}\|^2 = C^2 \, \|A^{-1}|b\rangle\|^2$$

$\|A\| = \lambda_{N-1}$ ← maximum eigenvalue

$\|A^{-1}\| = \dfrac{1}{\lambda_0}$

$$\kappa(A) = \frac{\lambda_{N-1}}{\lambda_0} = \lambda_{N-1} C^{-1} \Rightarrow C^2 = \left(\frac{\lambda_{N-1}}{\kappa_A}\right)^2$$

$$\|A^{-1}|b\rangle\| \geq \frac{1}{\|A\|} \cdot \underset{1}{\underset{\shortparallel}{\||b\rangle\|}} = \frac{1}{\lambda_{N-1}}$$

$$\text{prob}(1) = C^2 \cdot \|A^{-1}|b\rangle\|^2 \geq \left(\frac{\cancel{\lambda_{N-1}}}{\kappa(A)}\right)^2 \cdot \frac{1}{\cancel{\lambda_{N-1}^2}} =$$

$$\Omega\left(\kappa(A)^{-2}\right)$$

In the worst case scenario we need to repeat $O\left(\kappa(A)^2\right)$ times the algorithm to obtain outcome 1

The gate complexity is dominated by the cost of QPE

To solve QLSP to precision ε : $\| |\tilde{x}\rangle - |x\rangle \| < \varepsilon$

We need to apply QPE $O(\kappa(A)/\varepsilon)$ times

each application of QPE has a cost which depends on the gate complexity for encoding $U = e^{2\pi i A}$

We have exponential advantage if U can be constructed with a circuit with depth

$O(\mathrm{polylog}(N))$

$$N s\, \kappa(A) \log \frac{1}{\varepsilon} \to \mathrm{polylog}(N)\, \kappa(A) \frac{1}{\varepsilon}$$

Ewing Tong.